Epidemiology
Week 3
Cohort Study

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1 Definition and Method

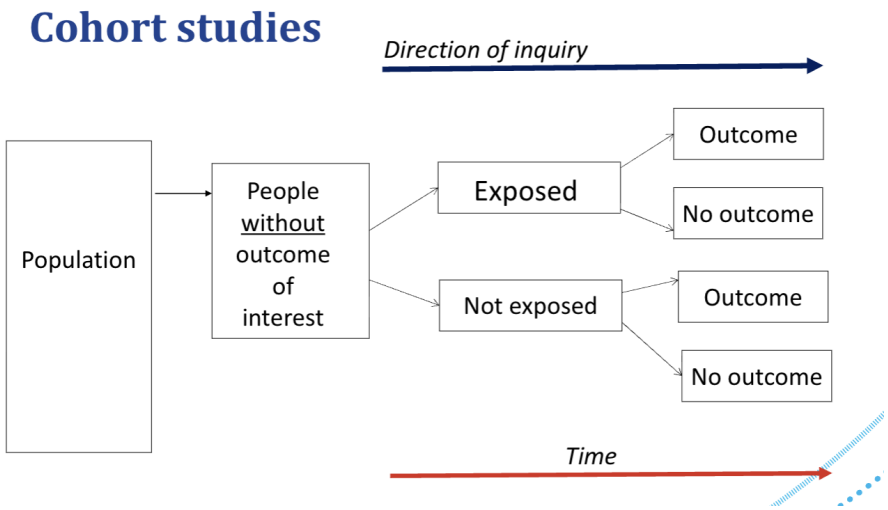


Figure 1: Summary of the three studies differences

As the image above says, starting from choosing people **WITHOUT** the outcome of interest the study population then follow-up splitting them into either the exposed or not exposed group and ending up with the outcome or no outcome from both groups.
Conducting a cohort study follows these steps:

1. Research question - Think about how exposure is associated with the outcome(s)

2. Identify the study population - This depends on how common the exposure of interest is
3. Measure the exposure
4. Follow-up participants and determine their outcome status
5. Data analysis
6. Interpret the results
7. Assess possible sources of error

Measure the exposure

Measuring the exposure can be gathered from a number of sources:

- Medical records / tests
- Measures of blood
- Physical measurements, e.g. height, weight, etc.
- Direct questioning to the subject of interest, questionnaires
- Environmental measures, e.g. water sampling

It is highly important to define the exposure, such as *how has epxosure been ascertained?* or *how often has exposure been measured?* to avoid boundary (a maybe situation) cases. Confounding variables also need to be measured.

Follow up participants and determine their outcome status

Just like defining the exposure, outcome also needs to be defined clearly and its ascertainment for the same reason to avoid boundary cases. Furthermore, things like *do all the population have similar risk of outcome?* or whether the population at risk has changed over time needs to be thought out thoroughly.

Data analysis

Cohort study calculates **incidence** as a risk or rate, which can then be compared to get the risk / rate ratio. The difference in calculating risk or rate is typically on whether the follow-up time is identical (or nearly) for the cohort members, risk is for when the follow-up time is identical, while rate is when there's variation between the follow-up time of the cohort members.

Analysis should also allow for the changing nature of exposure and for the differences in groups (by using standardisation, multivariate adjustments, Poisson regression, or Cox regression). One thing to note is **Effect modification**, it is when the effect of an exposure or treatment of an outcome changes depending on the level of a third variable, i.e. *A third variable distorts the exposure, outcome, or both*

Calculating risk

When the follow-up time is **identical** (or nearly) identical for all cohort members.

	outcome		
exposure	Yes	No	Total
exposed	a	b	a+b
not exposed	c	d	c+d

Figure 2: Calculate risk

The formulas for the risks are:

Risk in exposed, $r_e = \frac{a}{a+b}$	**Risk in unexposed**, $r_0 = \frac{c}{c+d}$
Risk ratio $= \frac{r_e}{r_0}$	**Risk difference** $= r_e - r_0$

Table 1: Calculating risk in cohort studies

Calculating rate

When there is **variation** between cohort members in the duration of follow-up time.

	outcome		
exposure	Yes	No	person-time
exposed	a	b	y_e
not exposed	c	d	y_0

Figure 3: Calculate rate

The formulas for the rates are:

Rate in exposed, $r_e = \frac{a}{y_e}$	**Risk in unexposed**, $r_0 = \frac{c}{y_0}$
Risk ratio $= \frac{r_e}{r_0}$	**Risk difference** $= r_e - r_0$

Table 2: Calculating rate in cohort studies

Interpret results

Interpreting the results follows these 3 principles:

- RR = 1 -> No association
- RR > 1 -> Risk in exposed is greater than risk in not exposed
- RR < 1 -> Risk in exposed is less than risk in not exposed

Where RR is the risk / rate ratio

Assess possible sources of error

Errors in studies can come from multiple sources, but primarily they may come from bias, confounding variables, or even chance. Examples of bias are information bias or attrition bias that came due to cohort members not available for the follow-up. It is highly important to collect information about possible confounders.

Pros and cons of cohort study

Advantage	**Disadvantage**
Temporal sequence of events can be established	Slow and usually expensive
Rare exposures can be studied	Inefficient for **rare disease**
Can examine multiple outcomes	Differential loss to follow up can lead to attrition /information bias
Can provide information about the time course of outcome (natural history)	Exposure stats can change
Can measure incidence rate directly	Can become historical

Table 3: Pros and cons of cohort study

2 Types of Cohort Studies